

Samitha Bandara

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GitHub | LinkedIn | Sri Lanka

Research interests: deep learning, reinforcement learning, robotics, and computer vision.

Education

University of Moratuwa, Sri Lanka

BSc Engineering Honours in Electronic and Telecommunication Engineering

Final-year undergraduate | Expected completion: 31 May 2027

CGPA: **3.82 / 4.00** | Dean's List: Semesters 1, 2, and 3.

Nalanda College, Colombo

GCE Advanced Level, Physical Science (2022): 3 A grades; island rank 278.

Research Experience

Research Assistant - Singapore Management University

School of Computing and Information Systems

- Researched efficient vision-language models through visual information reuse and compression.
- Evaluated multimodal models on visual question-answering benchmarks.

Supported through SMART M3S. Supervised by Prof. Archan Misra, with guidance from Dr. Dulanga Weerakoon.

Two manuscripts under review: first author on one, co-author on the other.

Ongoing Research

Point-cloud vision-language models

Studying how models can reuse visual information when a 3D scene changes.

Aerial robot stabilization

Using reinforcement learning to help a drone with a robotic arm stay stable in simulation.

Final-year project in collaboration with Prof. Rajat Talak, National University of Singapore.

Selected Projects

Transformer Optimization

Adapters, quantization, KV caching, and efficiency techniques for transformer models.

GitHub

CoreLlama

A Llama 2 implementation from scratch in PyTorch.

GitHub

VLM-gamma

A vision-language model built from scratch with a Vision Transformer and transformer decoder.

GitHub

gpt2-lite

A GPT-2 reproduction with a custom training pipeline and BPE tokenization.

GitHub

Technical Skills

Programming

Python, C, C++, MATLAB

Machine learning

PyTorch, Hugging Face, NumPy, OpenCV

Robotics

Reinforcement learning, Isaac Sim, Isaac Lab, Gazebo, PPO

Systems

Linux, Slurm, Docker, Git

Embedded systems

STM32, embedded C, PCB design, sensors